



SYDNEY
BOYS
HIGH
SCHOOL

2019

YEAR 9 YEARLY
MATHEMATICS

General Instructions:

- All questions may be attempted.
- Write using black or blue pen.
- Marks may be deducted for careless or badly arranged work
This can include insufficient working out.
- All working and answers are to be written in this test booklet.
- If you wish to rewrite an answer, draw a line through your faulty answer and rewrite your answer on pages 26 and 27 of this booklet.
Show the number and part of the answer being rewritten and write “See Back” at the original question.
- NESA approved calculators may be used.
- Clearly indicate your class by placing an **X** next to your class.

Time Allowed: 90 minutes

Examiner: AW

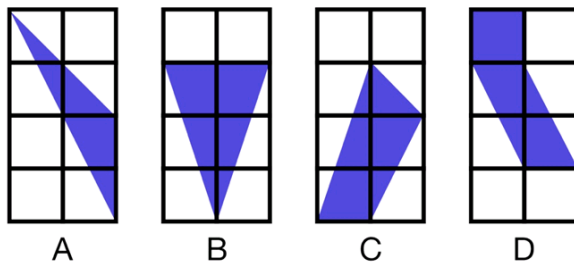
Name: _____

Class	Teacher	
9MaA	Mr A Wang	
9MaB	Mr James	
9MaC	Ms H Chan	
9MaP	Mr Fuller	
9MaL	Ms J Chan	
9MaU	Ms Kilmore	
9MaS	Ms Millar/Mr R Wang	

Question	Marks
1	/ 16
2	/ 17
3	/ 16
4	/ 16
5	/ 17
6	/ 18
Total	/ 100

Question 1 (16 marks)

- a) Circle the letter of the polygon that does not have the same area as the others. 1



- b) Expand then collect like terms for the following expressions:

(i) $4(a+1) - 6a - 5$ 1

(ii) $(2a+b)(a-11b)$ 1

- c) JoJay, a casual labourer, is paid \$21.78 per hour. How much will he earn if he works a 35 hour week? 1

- d) Show by substitution that $a = 2$ is the solution to the following equation. 1

$$11(a + 8) = 5(a + 20)$$

- e) Write $2500 \times 3\,640\,000$ in scientific notation. 1

Question 1 Continued

f) Solve the following equations, giving answers in simplest exact form.

(i) $4p + 6 = 37$

1

(ii) $\frac{y}{5} + 3 = 7$

1

(iii) $2y^2 + 9 = 23$

2

(iv) $\frac{2x-1}{4} = \frac{3x+2}{8}$

2

Question 1 Continued

g) Express $5x^{-3}$ with a positive index. **1**

h) Find the gradient m and y -intercept of these lines:

(i) $y = 3x + 2$ **1**

(ii) $4x + 3y + 6 = 0$ **2**

BLANK PAGE

Question 2 (17 marks)

- a)** PaPa added 10% GST onto the price of a book valued at \$29.90 to get its retail price. He then discounted the retail price, by 10% to get the 'on sale' price.

(i) What was the retail price? **1**

(ii) What is the 'on sale' price? **1**

(iii) By what percentage should PaPa have discounted the retail price to get back to what the book was valued at? **2**

Question 2 Continued

b) Simplify the following expressions, leaving each with only positive indices:

(i) $x^5 \times x^7$ **1**

(ii) $p^{-\frac{1}{2}} \times p^{\frac{1}{3}}$ **2**

(iii) $\frac{6a^5b^6}{4a^3b^2}$ **1**

(iv) $\frac{2x^2 + 6}{8x^6 + 24x^4}$ **2**

Question 2 Continued

c) A line goes through the points $A(1, 3)$ and $B(-3, 4)$.

(i) Find the midpoint of the interval joining A and B . **1**

(ii) Find the gradient of the straight line through A and B . **1**

(iii) Find the exact distance from A to B . **1**

(iv) What is the equation of the straight line passing through both A and B ? **2**
Leave your answer in general form.

(v) Find the equation of the line passing through $(5, 7)$ and perpendicular to the line passing through A and B . **2**

BLANK PAGE

Question 3 (16 marks)

a) Simplify $(\sqrt{5} - 3)(\sqrt{5} + 3)$. **1**

b) Solve the following inequality and show the solution on a number line. **2**
 $17 - 2m \leq 16$

c) Make x the subject of the formula below, leaving your answer as a simplified fraction. **2**
$$\frac{x+y}{y} = \frac{y}{a} + \frac{a}{y}$$

Question 3 Continued

d) Solve for x .

2

$$\frac{4x-1}{3} - \frac{2x+3}{4} = \frac{x+1}{6}$$

e) Fully factorise the following expressions:

(i) $100x^2 - 81y^2$

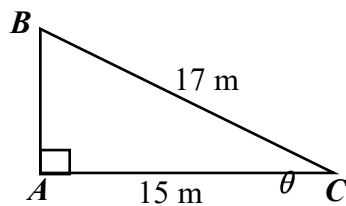
1

(ii) $x^2 - 9x - 36$

1

Question 3 Continued

f) With respect to the diagram below, write down the value of:

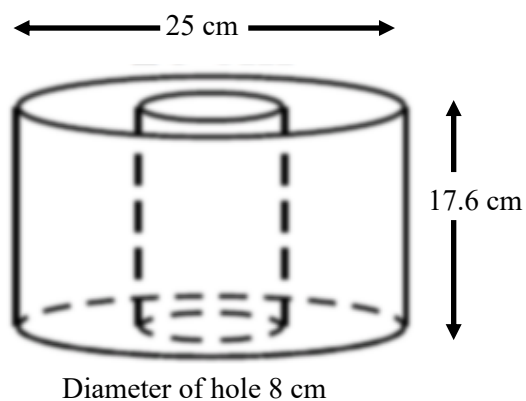


(i) $\cos \theta$ 1

(ii) $\tan \theta$ 2

(iii) θ (nearest minute) 1

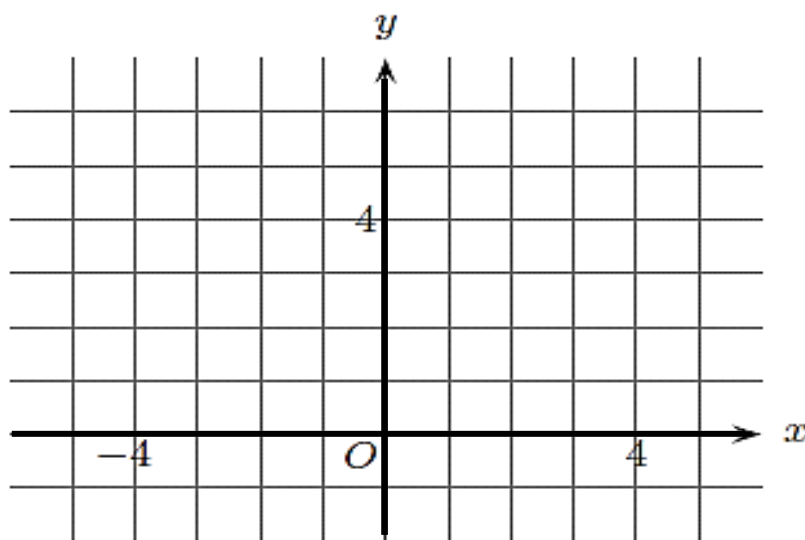
g) Find the surface area of this solid correct to one decimal place. 3



BLANK PAGE

Question 4 (16 marks)

- a) By using graphical means, find the point of intersection of $x + 3y = 1$ and $y = 2x + 5$. 3



- b) Fully factorise the following expressions:

(i) $12x^2 - 23x + 5$ 2

(ii) $x^3 - 25x - 3x^2 + 75$ 2

Question 4 Continued

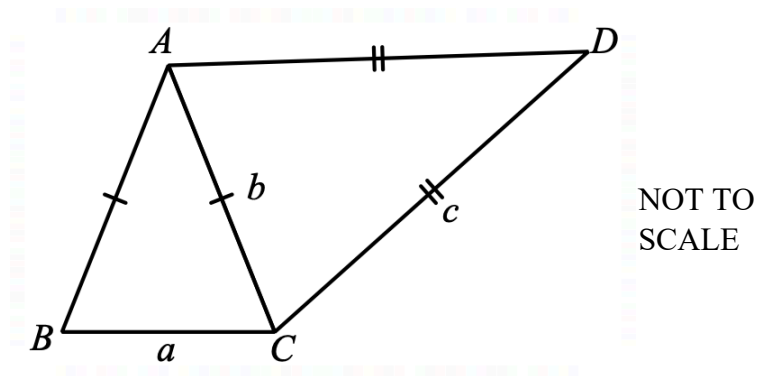
- c) Express $\frac{2\sqrt{3}}{\sqrt{3}-1}$ with a rational denominator. **2**

- d) A packet of lollies contains 5 red lollies and 15 green lollies. Two lollies are selected at random without replacement. By drawing a tree diagram, or otherwise, find the probability that the two lollies are different colours. **3**

Question 4 Continued

- e) In the diagram below, both $\triangle ABC$ and $\triangle DAC$ are isosceles triangles.

$AB = AC$, $AD = CD$ and $\angle BAC = \angle ADC$.



- (i) Prove that $\triangle ABC \sim \triangle DAC$.

2

- (ii) Hence prove that $b^2 = ac$.

2

BLANK PAGE

Question 5 (17 marks)

- a) Consider the 2019-2020 yearly tax table below.

Mrs West earns \$42 042 dollars per year, and \$567 income from other sources. She has deductions of \$665, and has paid PAYE (Pay As You Earn) tax instalments totalling \$4042.

With the help of the table, find:

Taxable income	Tax on this income
0 – \$18,200	Nil
\$18,201 – \$37,000	19c for each \$1 over \$18,200
\$37,001 – \$90,000	\$3,572 plus 32.5c for each \$1 over \$37,000
\$90,001 – \$180,000	\$20,797 plus 37c for each \$1 over \$90,000
\$180,001 and over	\$54,097 plus 45c for each \$1 over \$180,000

- (i) her taxable income.

1

- (ii) the tax payable on her taxable income.

2

The above rates **do not** include the Medicare levy of 2%.

- (iii) the refund, or balance payable, after the Medicare levy of 2% is paid.

2

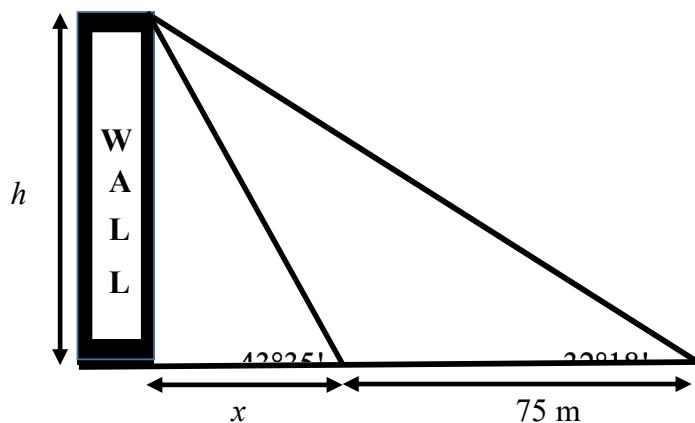
- b) Express as a single fraction, with a rational denominator:

$$\sin(60^\circ) + \tan(30^\circ) + \cos(0^\circ)$$

2

Question 5 Continued

- c) Consider the diagram below, showing a wall supported by two stays.



- (i) Find two expressions for the height of the wall h , in terms of x .

2

- (ii) Solve for x .

1

- (iii) Find the height of the wall.

1

Question 5 Continued

d) Simplify:

3

$$\frac{x^2 - 25}{3x^2 + 15x} - \frac{x^2 + 4x - 5}{x^2 - x}$$

e) Over 1 year you want to invest \$30 000.

3

Part of this will be invested in a 5% p.a. simple interest rate account.

The remainder will be invested in the family company returning 7% p.a (simple interest) on the investment.

What is the least you can invest in the family company and still get at least \$1900 interest?

(Hint: construct an inequality where x is the amount invested in the family company.)

BLANK PAGE

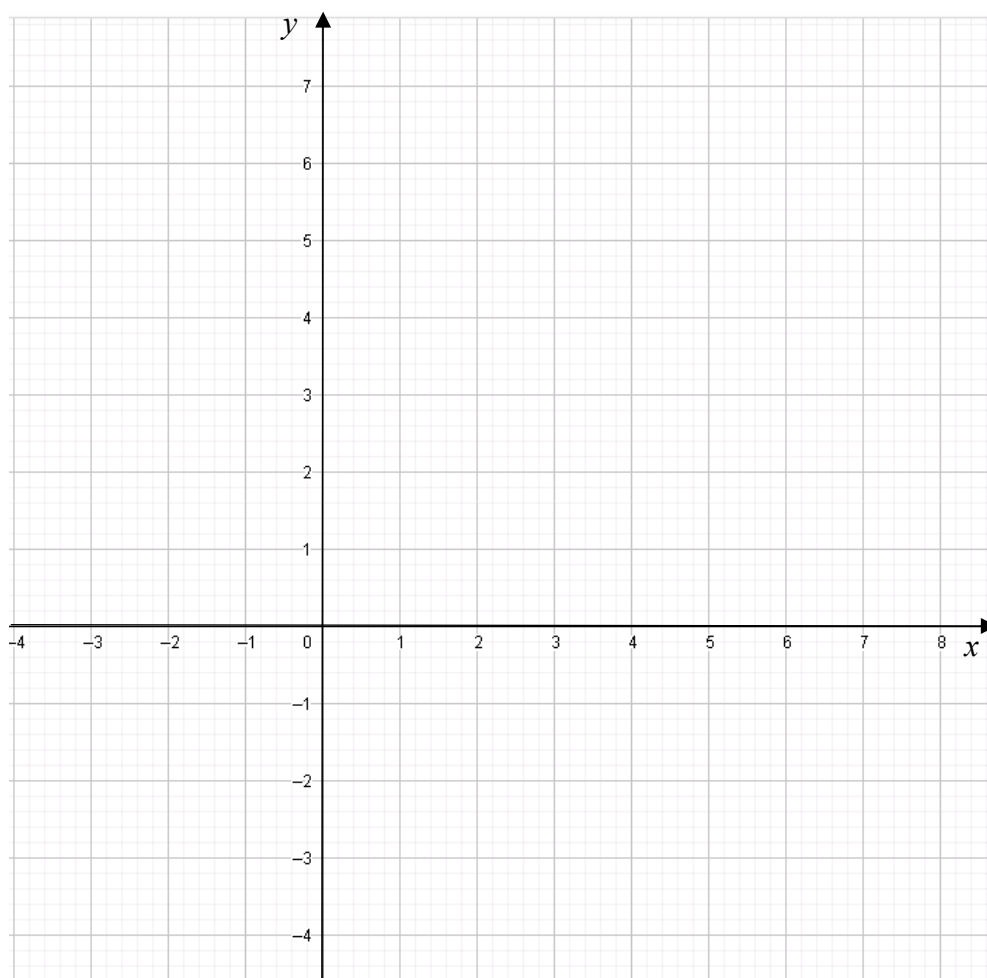
Question 6 (18 marks)

a) On the axes below graph and label the following lines:

(i) $y = -1$ 1

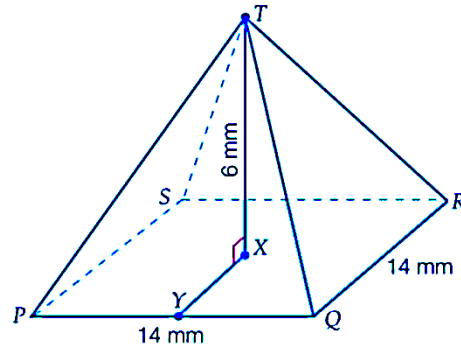
(ii) $y = \frac{2x}{3}$ 1

(iii) $y + 2x = 6$ 2



Question 6 Continued

b)



In the diagram above $PQRST$ is a right pyramid, Y is the midpoint of PQ and X is the centre of the square base $PQRS$.

(i) Show that $\angle TYX = 40^\circ 36'$ (nearest minute).

1

(ii) Show that $\angle TQX = 31^\circ 13'$ (nearest minute).

3

c) Which number's square root is equal to two times its cube root?

2

Question 6 Continued

- d)** A game involves rolling two six-sided dice, followed by rolling a third six-sided die. To win the game, the number rolled on the third die must lie between the two numbers rolled previously. For example, if the first two dice show 1 and 4, the game can only be won by rolling a 2 or 3 with the third die.

(i) What is the probability that a player has no chance of winning before rolling the third die? **2**

(ii) What is the probability that the player wins the game? **2**

Question 6 Continued

- e) A sequence is *arithmetic* if the difference between any two consecutive terms is always the same i.e. constant.

This means that if the n numbers $a_1, a_2, a_3, \dots, a_{n-1}, a_n$ form an arithmetic sequence then $a_2 - a_1 = a_3 - a_2 = \dots = a_n - a_{n-1} = d$, for some constant d .

- (i) Find an expression for $a_n - a_1$ in terms of n and d .

1

- (ii) If $\frac{1}{x_1}, \frac{1}{x_2}, \frac{1}{x_3}, \dots, \frac{1}{x_n}$ form an arithmetic sequence, show that

3

$$x_1x_2 + x_2x_3 + x_3x_4 + \dots + x_{n-1}x_n = (n-1)x_1x_n.$$

End of paper